

LUNAR BUILD MISSION PLAN — South-Pole Site Development
Moon · 2026-06-03 · cut-fill balanced, optimized sequence

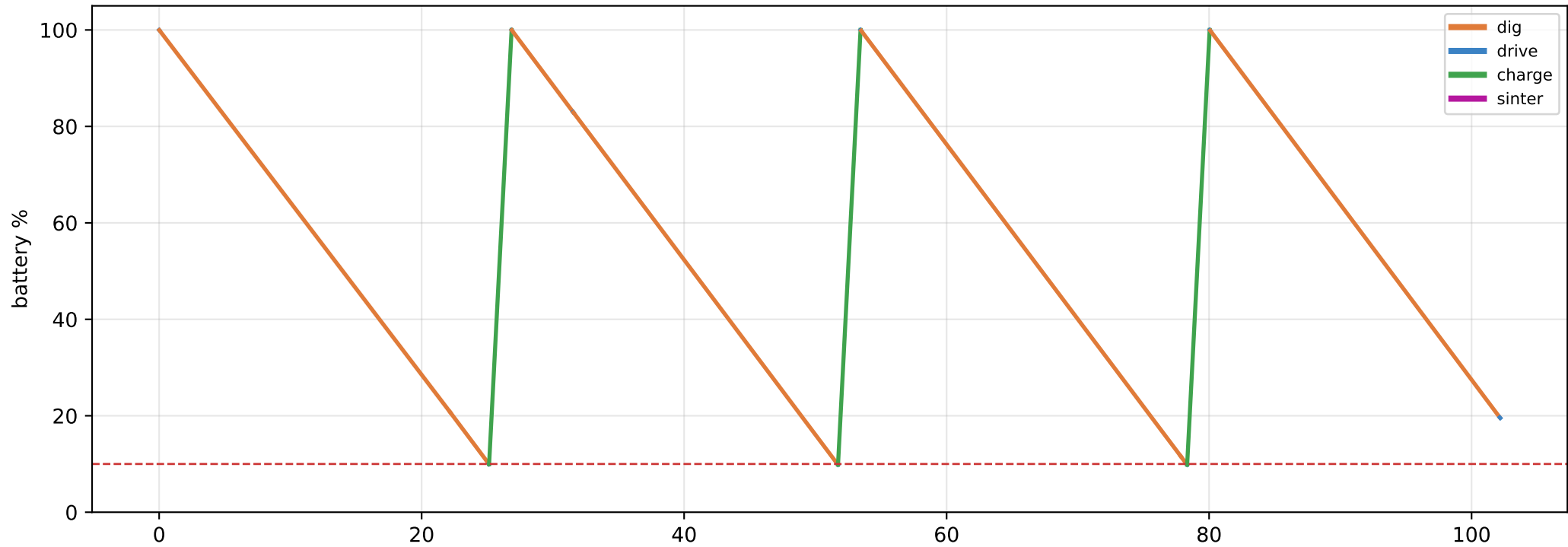
#	Trip	kind	Site (x,y)	Mass t	Duration	Energy (chg)
1	Grade access road → Fill crater di	cutfill	(15,5)	0.83	22.2 h	0.8
2	Excavate spoil: Grade access road	dig	(15,5)	0.32	9.4 h	0.3
3	Level landing pad → Build blast be	cutfill	(40,30)	1.82	46.3 h	1.6
4	Excavate spoil: Level landing pad	dig	(40,30)	0.94	24.3 h	0.8

MATERIAL BALANCE
cut 3.9 t → fill 2.7 t · surplus(spoil) 1.3 t · deficit(import) 0.0 t · sinter 0.00 t

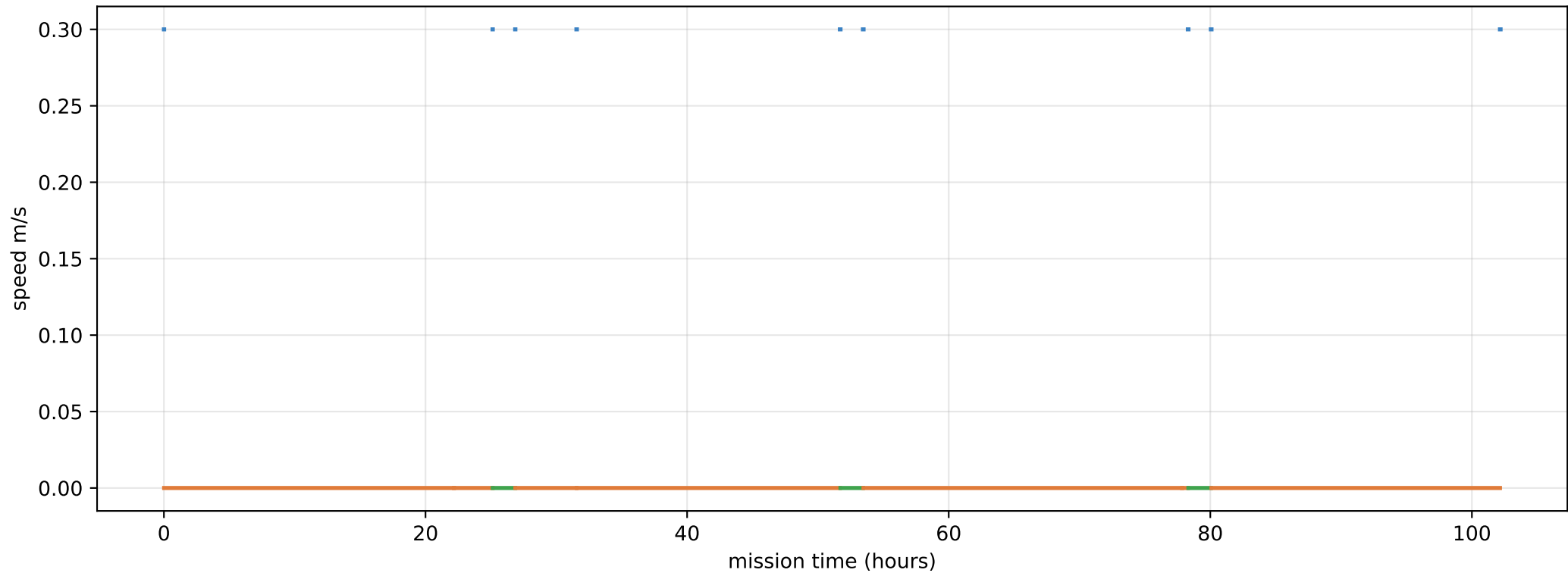
TOTALS project 4.3 d (102 h) moved 3.9 t (131 drum loads)
energy 16.8 MJ (3.5 charges, 3 recharge stops) drive 4.09 km
89 drum cycles; fill SENSED from motor current +/-2.6% (>half full), no load cell
per-sortie range 25 km (slope+slip @ 17 deg; 32 km flat)
1 lunar day ~ 30 Earth-days (354 h light)
ConOps: 70 km + 5-10 t / 11 d -> drive ~2.0 packs vs dig ~4-9 packs: drums dominate

Cut-fill balanced (excavated material routed to nearest fill; surplus→spoil, deficit→import). Grounded: per-body density/gravity (bodies.json); IPEx — 0.30 m/s, 42 kg/hr dig, 4151 J/kg, 135 J/m (slip-adjusted on a DEM), 4.79 MJ battery, 30 kg/drum; SINTER 0.92 MJ/kg [CALIB] (~220× dig). Recharge 700 W + sinter-head 1000 W are [CALIB]. Pluggable sequencer × objective; battery-aware mid-task recharge.

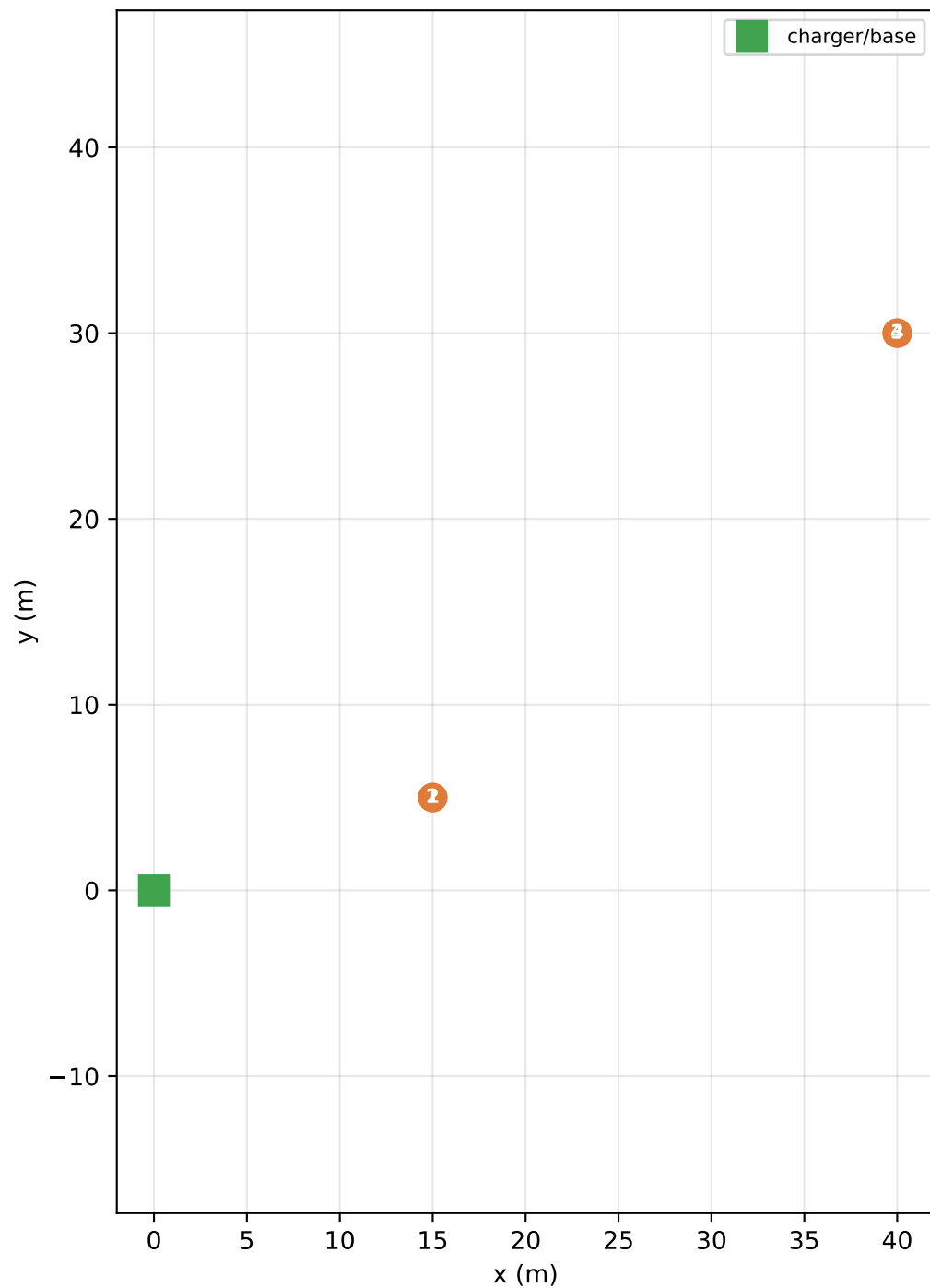
Battery draw over the planned project



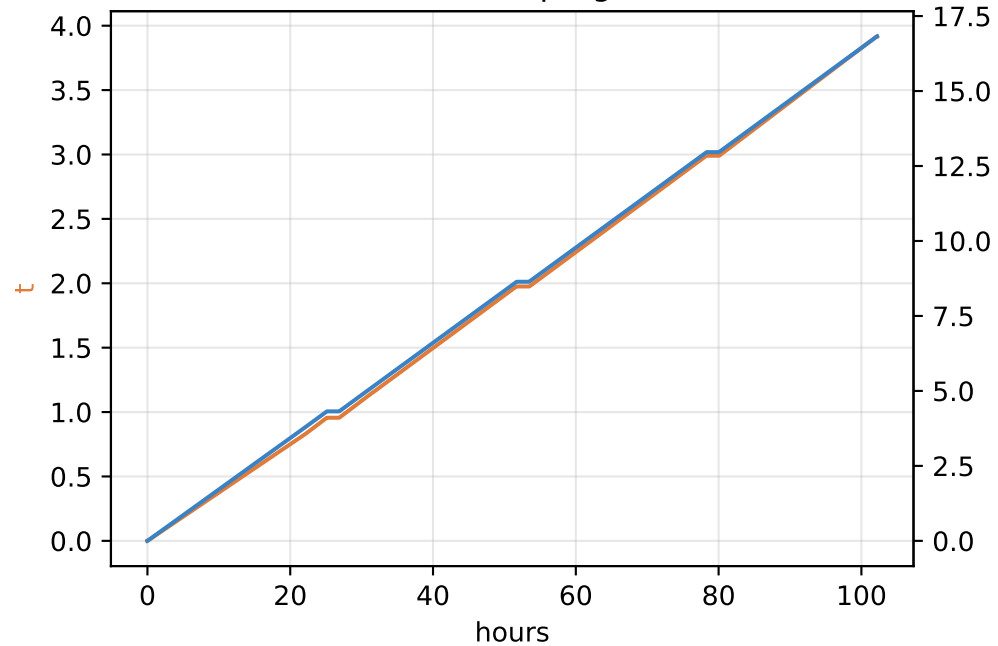
Speed profile



Site route + material flows (cut→fill)



Cumulative progress



Energy per trip (battery charges)

